

Last name _____

First name _____

LARSON—MATH 756—HOMEWORK WORKSHEET 07
Test 1 Review

Be verbose.

Definitions. State and provide an example.

1. If M is a matching in a graph G , what is an M -*alternating* path in G ?
2. If M is a matching in a graph G , what is an M -*augmenting* path in G ?
3. What is a Vertex Packing Integer Program?
4. What is the *fractional independence number* α_f ?
5. What is *Lovász' theta function* ϑ .

Theorems

6. State and prove Berge's Theorem.
7. State and prove Hall's Theorem.
8. State König's Theorem.

Problems

9. What is an algorithm for finding a maximum independent set in an arbitrary graph?
10. How many iterations will the naive independence number algorithm for a graph with order n require?
11. Find a formula for α for a cycle with n vertices.
12. (Gallai Identity) Prove, for any graph, that $\alpha + \beta = n$, where β is the vertex covering number and n is the order of the graph.
13. Why is $\alpha \leq \alpha_f$ for any graph?

(over)

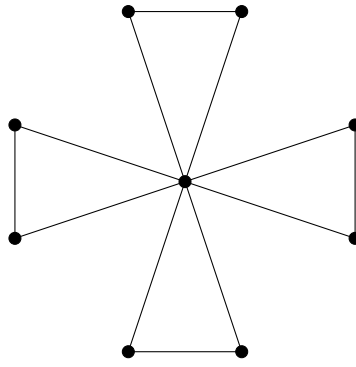


Figure 1: A flower F_4 with four petals.

14. Write the Vertex Packing IP model for the graph F_4 .
15. Find an optimal feasible solution for the VIP, and argue that it is optimal.
16. Write the Vertex Packing LP model for the graph F_4 .
17. Find an optimal feasible solution for the VPLP, and argue that it is optimal.
18. Add *clique constraints* to your VPLP: add one additional constraint for each of the four triangles.
19. Find an optimal feasible solution for the updated VPLP, and argue that it is optimal.

Orthonormal Representations and Lovász Theta

20. Find any orthonormal representation $\{\hat{x}_i\}$ for F_4 . Explain.
21. Find three different unit vectors $\hat{c}$ and calculate $\max_i \frac{1}{(\hat{c}^T \hat{x}_i)^2}$ for each $\hat{c}$.
22. Find a unit vector $\vec{c}$ and orthonormal representation $\{\hat{x}_i\}$ that minimizes $\max_i \frac{1}{(\vec{c}^T \hat{x}_i)^2}$ for the graph F_4 .